

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1019

Lawrence Hill 100 kV
December 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1019. The Interconnection Customer proposes a 538.8 MW natural gas combined cycle facility interconnecting at the Lawrence Hill 100 kV substation in the Illinois area. The request was received on July 03, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 350 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,849,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: N VLY to LAWRENCE HILL 100kV L
Contingency: TERRYVILLE to LAWRENCE HILL 100kV
Loading: 134.6% (Rating: 135 MVA)
DFAX: 43.4%
Pre-project loading: 38.6%
2. Facility: CANADYS STEAM SUBSTATION to LAWRENCE HILL 100kV Line 2
Contingency: CANADYS STEAM to LAWRENCE HILL 100kV
Loading: 125.8% (Rating: 147 MVA)
DFAX: 15.0%
Pre-project loading: 71.0%
3. Facility: DARLINGTON COUNTY to CERES 100kV L
Contingency: LAWRENCE HILL to N VLY 100kV
Loading: 124.2% (Rating: 105 MVA)
DFAX: 14.7%
Pre-project loading: 48.7%
4. Facility: LAWRENCE HILL to CANADYS STEAM 100kV Cir. 1
Contingency: CANADYS STEAM to LAWRENCE HILL 100kV Cir. 2
Loading: 122.0% (Rating: 152 MVA)
DFAX: 15.0%
Pre-project loading: 68.8%
5. Facility: MUSKOGEE SUBSTATION to DARLINGTON ROD 100kV
Contingency: RICHLAND to DEKALB 500kV
Loading: 111.1% (Rating: 515 MVA)
DFAX: 6.7%
Pre-project loading: 104.1%

6. Facility: CHAMBERLIN to INVOLTA 230kV
Contingency: HYLEBOS to RAVENNA 500kV
Loading: 110.0% (Rating: 500 MVA)
DFAX: 5.3%
Pre-project loading: 104.2%
7. Facility: BARAGA VLG to GRAY'S FLAT 230kV L
Contingency: MADELINE to E METHUEN 230kV
Loading: 109.7% (Rating: 1,571 MVA)
DFAX: 8.9%
Pre-project loading: 106.6%
8. Facility: E METHUEN 500 kV/230 kV XFMR
Contingency: BEAVER to TREGO 500kV line
Loading: 109.7% (Rating: 2,846 MVA)
DFAX: 16.0%
Pre-project loading: 106.7%
9. Facility: GRAY'S FLAT to E METHUEN 230kV line
Contingency: MADELINE to E METHUEN 230kV
Loading: 109.4% (Rating: 1,589 MVA)
DFAX: 8.9%
Pre-project loading: 106.4%
10. Facility: GRUVER to GALEVILLE 230kV
Contingency: IMPERIAL SUBSTATION to EUFAULA 500kV
Loading: 109.3% (Rating: 791 MVA)
DFAX: 6.6%
Pre-project loading: 104.8%
11. Facility: MADELINE SUBSTATION to BARAGA VLG 230kV
Contingency: GRAY'S FLAT to EAST METHUEN 230kV
Loading: 108.8% (Rating: 1,044 MVA)
DFAX: 6.5%
Pre-project loading: 105.4%
12. Facility: NEWPORT TIE to CANAL RAD 230kV
Contingency: BEAVER to TREGO 500kV
Loading: 108.3% (Rating: 1,484 MVA)
DFAX: 10.4%
Pre-project loading: 104.5%
13. Facility: E BELLEVILLE to DARLINGTON COUNTY 100kV
Contingency: ALSIP to CRYSTAL VALLEY 100kV
Loading: 107.6% (Rating: 153 MVA)
DFAX: 7.9%
Pre-project loading: 79.6%
14. Facility: EUFAULA to IMPERIAL 500kV line
Contingency: TREGO to GALION GENERATING STATION 500kV Line
Loading: 107.5% (Rating: 4,273 MVA)
DFAX: 29.1%
Pre-project loading: 103.8%
15. Facility: BURRIS PIT to HEBO 230kV
Contingency: TREGO to BEAVER 500kV
Loading: 107.5% (Rating: 1,482 MVA)
DFAX: 7.9%
Pre-project loading: 104.6%
16. Facility: WABASH 500/230kV Transformer Bank
Contingency: DEKALB to RICHLAND 500kV

- Loading: 107.4% (Rating: 1,348 MVA)
DFAX: 19.5%
Pre-project loading: 99.6%
17. Facility: MONFORT to WABASH 230kV
Contingency: DEKALB to RICHLAND 500kV
Loading: 106.8% (Rating: 1,296 MVA)
DFAX: 18.2%
Pre-project loading: 99.3%
18. Facility: WAYNE DALE to MONTPELIER ELECTRIC GENERATING STA 230kV
Contingency: RICHLAND to DEKALB 500kV
Loading: 106.3% (Rating: 1,036 MVA)
DFAX: 21.0%
Pre-project loading: 95.4%
19. Facility: CARLEY ROAD SUBSTATION to JAMES DE YOUNG 230kV Line
Contingency: E METHUEN 500-230 kV XFMR
Loading: 106.2% (Rating: 1,209 MVA)
DFAX: 7.1%
Pre-project loading: 103.0%
20. Facility: GALION GENERATING STATION 500/230 kV Transformer Bank
Contingency: TREGO to BEAVER 500kV
Loading: 106.0% (Rating: 1,792 MVA)
DFAX: 10.8%
Pre-project loading: 102.7%
21. Facility: ADIRONDACK to JAMES DE YOUNG 230kV Line
Contingency: ST. LAWRENCE to JAMES DE YOUNG 230kV L
Loading: 105.7% (Rating: 1,366 MVA)
DFAX: 7.4%
Pre-project loading: 102.8%
22. Facility: CRYSTAL VLY SUBSTATION to ALSIP 100kV
Contingency: DARLINGTON COUNTY to EBELLEVILLE 100kV
Loading: 105.6% (Rating: 141 MVA)
DFAX: 8.9%
Pre-project loading: 71.5%
23. Facility: CANADYS STEAM to CRESTWOOD 100kV
Contingency: CRESTWOOD SUBSTATION to CANADYS STEAM 100kV Cir. 2
Loading: 105.6% (Rating: 146 MVA)
DFAX: 9.7%
Pre-project loading: 70.0%
24. Facility: CANADYS STEAM SUBSTATION to CRESTWOOD 100kV Cir. 2
Contingency: CANADYS STEAM to CRESTWOOD 100kV #1
Loading: 105.6% (Rating: 146 MVA)
DFAX: 9.7%
Pre-project loading: 70.0%
25. Facility: TREGO to GALION GENERATING STATION 500kV
Contingency: TREGO to BEAVER 500kV
Loading: 105.4% (Rating: 3,288 MVA)
DFAX: 20.6%
Pre-project loading: 102.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N VLY to LAWRENCE HILL 100kV L	Rebuild line with higher-capacity conductor	\$1,885,000	81% (\$1,533,000)
CANADYS STEAM SUBSTATION to LAWRENCE HILL 100kV Line 2	Rebuild line with higher-capacity conductor	\$10,480,000	84% (\$8,849,000)
DARLINGTON COUNTY to CERES 100kV	Rebuild line with higher-capacity conductor	\$1,016,000	45% (\$457,000)
CANADYS STEAM to LAWRENCE HILL 100kV #1	Rebuild line with higher-capacity conductor	\$8,217,000	84% (\$6,938,000)
MUSKOGEE to DARLINGTON ROAD SUBSTATION 100kV L	Rebuild line with higher-capacity conductor	\$17,000,000	4% (\$611,000)
CHAMBERLIN to INVOLTA 230kV	Reconductor line	\$4,178,000	4% (\$151,000)
BARAGA VLG to GRAY'S FLAT 230kV L	Reconductor line	\$9,077,000	3% (\$250,000)
E METHUEN 500/230kV XFMR	Replace transformer with higher-rated unit	\$35,691,000	3% (\$1,009,000)
GRAY'S FLAT to E METHUEN 230kV line	Reconductor line	\$11,515,000	3% (\$318,000)
GALEVILLE to GRUVR 230kV line	Reconductor line	\$4,597,000	3% (\$142,000)
MADELINE SUBSTATION to BARAGA VLG 230kV	Reconductor line	\$4,586,000	3% (\$126,000)
NEWPORT TIE to CANAL RAD 230kV	Reconductor line	\$7,141,000	3% (\$218,000)
E BELLEVILLE to DARLINGTON COUNTY 100kV	Reconductor line	\$389,000	34% (\$131,000)
IMPERIAL to EUFAULA SUBSTATION 500kV	Reconductor line	\$49,561,000	3% (\$1,536,000)
HEBO to BURRIS PIT 230kV L	Reconductor line	\$14,144,000	3% (\$399,000)
WABASH 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$25,178,000	3% (\$838,000)
MONFORT to WABASH 230kV line	Reconductor line	\$10,100,000	3% (\$337,000)
WAYNE DALE to MONTPELIER ELECTRIC GENERATING STA 230kV line	Reconductor line	\$20,238,000	3% (\$680,000)
CARLEY RD to JAMES DE YOUNG 230kV Line	Reconductor line	\$11,889,000	3% (\$330,000)
GALION GENERATING STATION 500/230 kV TX	Replace transformer with higher-rated unit	\$36,847,000	3% (\$1,147,000)
ADIRONDACK to JAMES DE YOUNG 230kV Line	Reconductor line	\$10,928,000	3% (\$309,000)
CRYSTAL VLY SUBSTATION to ALSIP 100kV	Reconductor line	\$622,000	52% (\$326,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CANADYS STEAM to CRESTWOOD 100kV Cir. 1	Reconductor line	\$754,000	77% (\$582,000)
CANADYS STEAM to CRESTWOOD 100kV Cir. 2	Reconductor line	\$728,000	77% (\$562,000)
TREGO to GALION GENERATING STATION 500kV L	Reconductor line	\$66,231,000	3% (\$2,070,000)
	Total allocated to project		\$29,849,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.